

CASTLE: Contamination-Aware Spatial Transcriptomics Latent Embedding

A single-dataset generative model for single-cell-resolution spatial transcriptomics that jointly handles segmentation error, intrinsic/niche disentanglement, and multi-task representation learning

Method design specification

August 21, 2026

Abstract

We specify **CASTLE**, a probabilistic representation-learning model for single-cell-resolution imaging-based spatial transcriptomics (Xenium, CosMx, MERFISH/MERSCOPE) that operates on *one dataset at a time* with a *fixed gene panel*. CASTLE addresses three coupled problems that existing embedding methods treat separately or not at all. *(i) Segmentation error.* Observed counts are modelled as an exact mixture of a cell’s own transcripts, transcripts leaked from named neighbours through a *directional, mass-conserving, geometry-only* spillover kernel, and ambient background calibrated against negative-control probes. This yields per-cell-per-gene posterior source responsibilities, a decontaminated count matrix, and a calibrated segmentation-confidence score, all as by-products of fitting the embedding rather than as a separate preprocessing step. *(ii) Disentanglement.* Each cell receives a structured latent $Z_i = [z_i^c, z_i^n]$ in which z_i^n is computed by a rotation-invariant, multi-scale graph encoder that is architecturally forbidden from reading cell i ’s own expression, and the decoder is additive in log-space ($\log \tilde{\mu} = f^c(z^c) + f^n(z^n, z^c)$) so that the niche contribution $\Delta_{ig} = f_g^n$ is an explicit, per-cell, per-gene *spatial effect*. Identifiability is induced by an asymmetric information constraint that makes intrinsic capacity expensive and niche capacity cheap, together with a Hilbert–Schmidt independence penalty and a niche-sufficiency head. *(iii) One representation, four tasks.* A single fitted model supports cell-type clustering (on z^c), trajectory inference (on z^c , via a contamination-weighted diffusion operator and a local-isometry regulariser that prevents artificial cluster separation), niche/domain discovery (on the multi-scale z^n), and cell–cell communication — the last through a *spillover-aware null*: ligand–receptor co-expression is tested against synthetic data generated from the fitted model with the niche effect ablated but contamination, geometry and cell identity preserved, directly separating genuine signalling from segmentation artefact. We give the complete generative model, inference network, objective, optimisation and neighbour-sampling scheme, downstream estimators, default hyperparameters, an evaluation and ablation protocol with explicit decision thresholds, and an honest account of the model’s assumptions and failure modes.

Contents

1	Scope, and what changed relative to the earlier proposal	3
2	Notation and inputs	3
3	The three problems, stated precisely	4

4	Generative model	5
4.1	Latent variables and priors	5
4.2	Decoder: additive intrinsic-plus-niche log-expression	5
4.3	Observation model: exact three-source negative binomial mixture	6
4.4	Background from negative controls	7
4.5	Spillover kernel: geometry only, by design	7
4.6	Optional transcript-level refinement	8
4.7	Spatial doublets	8
5	Inference model	9
5.1	Unrolled purification encoder for z^c	9
5.2	Niche encoder for z^n : self-excluded, rotation-invariant, multi-scale	9
6	Objective	10
6.1	Graph-coupled ELBO	10
6.2	The asymmetric information constraint	10
6.3	Independence and sufficiency	11
6.4	Local isometry, for trajectory validity	11
6.5	Full objective	11
7	Optimisation and scalability	12
7.1	Two-hop neighbour sampling	12
7.2	Curriculum	12
7.3	Cost	12
8	Downstream: one model, four analyses	13
8.1	Cell type clustering	13
8.2	Trajectory inference	13
8.3	Niche and domain discovery	13
8.4	Cell–cell communication with a spillover-aware null	13
9	Uncertainty quantification	14
10	Hyperparameters and defaults	14
11	Evaluation, ablations, and decision thresholds	15
11.1	Ground-truth strategies	15
11.2	Baselines and metrics	15
11.3	Ablations	16
11.4	Decision thresholds (when to abandon parts of this design)	16
12	Assumptions, failure modes, and limitations	16
13	Implementation notes	17

1 Scope, and what changed relative to the earlier proposal

The earlier design (SPECTRA) was a cross-platform *foundation model*: protein-language-model gene tokens for panel-agnostic encoding, SwAV-style pretraining on a $\sim 50\text{M}$ -cell multi-platform corpus, and zero-shot transfer. That scope is now dropped. The revised requirement is a model that a group generating a Xenium/CosMx/MERFISH dataset fits *to their own dataset*, with a *fixed panel*, as a default first analysis.

This descopeing is not merely a subtraction; it changes the optimal design in four ways.

1. **The gene encoder can and should be panel-specific.** With a fixed panel of $G \in [100, 6000]$ genes, gene-identity embeddings are learned free parameters and the encoder is a plain MLP over a dense G -vector. There is no need for ESM2 tokens, set transformers over gene tokens, or mixture-of-experts protocol routing. This removes the single largest source of architectural complexity and the largest source of the “foundation models lose to PCA” risk.
2. **Model capacity can be spent on the physics of the assay instead of on transfer.** The freed capacity goes into an explicit, identifiable forward model of transcript misassignment, which the benchmark literature identifies as the dominant confounder of downstream cellular-state analysis in these platforms, and which no mainstream embedding method models.
3. **Technical covariates become nuisance variables within one experiment, not batch effects across experiments.** Field-of-view (FOV) index, z -plane, distance to tissue edge, and cell area are conditioned on directly through a conditional prior, rather than removed adversarially.
4. **Full-batch or near-full-batch inference is feasible.** A single slide is 10^5 – 10^6 cells. Neighbour-sampled minibatch training on one GPU is comfortable, which permits a graph-coupled likelihood (a cell’s likelihood depends on its neighbours’ latents) that would be prohibitive at 10^8 -cell atlas scale.

A note on “a single representation”. The specification asks for one representation serving clustering, trajectory, niche discovery, and communication. CASTLE produces one latent vector per cell, $Z_i = [z_i^c, z_i^n] \in \mathbb{R}^{d_c+d_n}$, from one model fitted with one objective. It is *structured*, not homogeneous: the two blocks are trained jointly and are mutually informative through the decoder, but they are constrained to carry different information. This is a deliberate design decision, and we state the reasoning plainly. An unstructured embedding cannot serve all four tasks, because the tasks make contradictory demands: clustering wants niche-invariant cell identity, niche discovery wants cell-identity-invariant context, and mixing the two is precisely the over-smoothing failure that degrades graph-neural-network embeddings on these data. Structuring the single latent is the minimal concession that makes the multi-task requirement coherent. Downstream tasks read the block they need; nothing is refitted.

2 Notation and inputs

Required inputs. A cell-by-gene count matrix X , centroids p_i , and cell areas/volumes a_i .

Strongly recommended. Segmentation masks or polygons (to compute c_{ij} and shape descriptors), nucleus masks (nucleus/cytoplasm ratio), and negative-control features (Xenium: negative control probes and negative control codewords; CosMx: negative probes and SystemControl features; MERFISH: blank barcodes).

Table 1: Symbols.

Symbol	Meaning
N, G	number of segmented cells; number of panel genes
$x_{ig} \in \mathbb{N}$	observed count of gene g assigned to cell i
$p_i \in \mathbb{R}^D$	cell centroid, $D \in \{2, 3\}$
a_i	cell area ($D=2$) or volume ($D=3$)
c_{ij}	length/area of the segmentation boundary shared by cells i and j
$d_{ij} = \ p_i - p_j\ $	centroid distance
$\mathcal{N}(i)$	spatial neighbours of i
u_i	observed technical covariates (FOV, z -plane, $\log a_i$, edge distance)
g_i	purely <i>geometric</i> morphology features of cell i (Sec. 4.5)
$z_i^c \in \mathbb{R}^{d_c}$	intrinsic latent
$z_i^n \in \mathbb{R}^{d_n}$	niche latent
$\mu_i \in \Delta^{G-1}$	cell i 's true (own) normalised expression profile
$\ell_i > 0$	cell i 's true total transcript content
$\eta_i \in [0, 1)$	fraction of i 's own transcripts <i>lost</i> to neighbours
$w_{j \rightarrow i}$	share of j 's leaked transcripts received by i ; $\sum_i w_{j \rightarrow i} = 1$
β_g, ν_i	background profile (from negative controls); per-cell background scale
ϕ_g	gene-specific inverse dispersion
λ_{ig}	expected observed count
π_{ig}^{own}	posterior probability an observed transcript is truly cell i 's
$\tilde{x}_{ig}$	decontaminated (purified) count
Δ_{ig}	spatial (niche) effect on log expression of gene g in cell i

Optional. The transcript-level table with per-molecule coordinates and quality scores, which activates the refinement module of Sec. 4.6; and z -coordinates, which activate the 3D leakage features.

Spatial graph. $\mathcal{N}(i)$ is built as the union of (a) the Delaunay triangulation with edges longer than the q -th percentile of edge length removed ($q = 99$ by default), and (b) a radius graph at R_1 . The union is used because Delaunay alone is unstable in sparse regions and radius/ k NN alone misrepresents anisotropic tissue. Edges are undirected for the niche encoder, and treated as directed for the spillover kernel. Sensitivity to this choice is an explicit ablation (Sec. 11).

3 The three problems, stated precisely

P1: segmentation error. Cell boundaries drawn in 2D (or from a nuclear expansion) on a 3D tissue section misassign transcripts. Three distinct error modes must be distinguished: (i) *spillover/bleed-through*: transcripts belonging to j assigned to i ; (ii) *ambient background*: transcripts belonging to no cell (or to optical/decoding error), detectable via negative-control features; (iii) *spatial doublets*: two real cells merged into one segment, or one cell overlapping another in z . Modes (i) and (ii) are additive contamination and are modelled generatively; mode (iii) is a *model-misspecification* event and is handled by detection and down-weighting (Sec. 4.7), not by pretending a single μ_i exists. The consequence of ignoring all three is that neighbouring-cell profiles are artificially correlated, which inflates exactly the quantities that niche and communication analyses measure.

P2: intrinsic vs. niche. We must separate “what this cell is” from “what surrounds it”. The difficulty is that the two are genuinely correlated in tissue (a T cell is usually next to other T cells),

so a niche encoder that reads the cell’s own expression, or that aggregates neighbours without constraint, will simply re-encode cell type — and a z^c estimated from own expression will silently absorb the niche’s causal effect on the cell. Without a structural asymmetry, the split is not identified.

P3: one representation, four tasks. Clustering rewards discrete, well-separated latents; trajectory inference requires a continuous manifold with meaningful local geometry; niche discovery requires context, not identity; communication requires *directed, gene-level* quantities, not a cell-level embedding at all. A representation that serves all four must therefore expose (a) an identity block, (b) a context block, and (c) a decoder whose parameters are themselves interpretable gene-level effects.

4 Generative model

4.1 Latent variables and priors

$$z_i^c \sim \mathcal{N}(m_\omega(u_i), \text{diag } s_\omega(u_i)), \quad (1)$$

$$z_i^n \sim \mathcal{N}(0, I_{d_n}). \quad (2)$$

The conditional prior (1) conditions the intrinsic latent on observed technical covariates u_i (one-hot FOV, z -plane, $\log a_i$, distance to tissue edge). Two reasons: it absorbs FOV/tile and depth artefacts without adversarial training, and, following the identifiable-VAE literature, an auxiliary-variable-conditioned prior from an exponential family is the standard sufficient condition for identifiability of the latent up to a permutation-and-scaling of coordinates. We use a diagonal Gaussian, i.e. a two-parameter exponential family per coordinate. m_ω, s_ω are small MLPs.

4.2 Decoder: additive intrinsic-plus-niche log-expression

$$\log \tilde{\mu}_{ig} = f_\theta^c(z_i^c)_g + \underbrace{f_\theta^n(z_i^n, z_i^c)_g}_{\Delta_{ig}}, \quad (3)$$

$$\mu_i = \text{softmax}(\log \tilde{\mu}_i) \in \Delta^{G-1}. \quad (4)$$

f^c is a standard MLP decoder. f^n is the *niche modulation*, and Δ_{ig} is by construction a per-cell, per-gene, log-scale spatial effect: the amount by which gene g in cell i is up- or down-regulated relative to what the cell’s intrinsic state alone would predict. Making the decomposition additive in log-space (rather than concatenating $[z^c, z^n]$ into a single MLP) is what makes the spatial effect *readable* rather than merely implicit; it is also what makes the communication analysis of Sec. 8.4 possible without a second model.

f^n takes z_i^c as a second argument because the same niche affects different cell types differently (context-dependent response is real biology), but it is regularised towards zero (Sec. 6) so that the intrinsic term explains as much as it can *subject to* the information constraint of Sec. 6.2.

Interpretable niche programs (optional but recommended). f^n may be given a masked-linear head over prior gene programs. Let $\mathcal{P}$ index curated ligand–receptor pairs and pathways, with binary masks $M_p \in \{0, 1\}^G$ restricted to genes present in the panel. Then

$$\Delta_{ig} = \sum_{p \in \mathcal{P}} M_{pg} v_{pg} [h_\theta(z_i^n, z_i^c)]_p + \underbrace{\sum_q r_{qg} [h_\theta(z_i^n, z_i^c)]_q}_{\text{de novo dims}}, \quad (5)$$

so that the first $|\mathcal{P}|$ niche coordinates are named program activities and the remainder are unconstrained, capturing structure absent from the prior knowledge base. With small panels most curated programs will be only partially covered; masks with fewer than 3 panel genes are dropped, and coverage is reported per program so users are not misled by a “pathway score” resting on one gene.

4.3 Observation model: exact three-source negative binomial mixture

Let ℓ_i be cell i ’s true total transcript content, with $\log \ell_i \sim \mathcal{N}(\log(\kappa_\ell a_i), \sigma_\ell^2)$ — a volume-linked prior that decouples “this cell is small” from “this cell is transcriptionally quiet”. The expected observed count is

$$\lambda_{ig} = \underbrace{(1 - \eta_i) \ell_i \mu_{ig}}_{\text{retained own}} + \underbrace{\sum_{j \in \mathcal{N}(i)} \eta_j \ell_j w_{j \rightarrow i} \mu_{jg}}_{\text{received spillover}} + \underbrace{\nu_i \beta_g}_{\text{ambient background}} \quad (6)$$

This parameterisation is *directional* and *mass-conserving*: cell j loses a fraction η_j of its content and that mass is distributed over j ’s neighbours by weights summing to one. Conservation is a hard constraint, not a penalty; it is what prevents the contamination term from acting as a free additive nuisance that absorbs biology.

The likelihood is constructed so that the mixture is *exactly* negative binomial and the source decomposition is available in closed form. Introduce a mean-one Gamma over-dispersion variable

$$\xi_{ig} \sim \text{Gamma}(\phi_g, \phi_g), \quad x_{ig} \mid \xi_{ig} \sim \text{Poisson}(\xi_{ig} \lambda_{ig}). \quad (7)$$

Proposition 1 (Exact NB marginal with closed-form source allocation). *Under (6)–(7), marginally $x_{ig} \sim \text{NegBinomial}(\lambda_{ig}, \phi_g)$ with $\mathbb{E}[x_{ig}] = \lambda_{ig}$ and $\text{Var}[x_{ig}] = \lambda_{ig} + \lambda_{ig}^2/\phi_g$. Moreover, writing $x_{ig} = x_{ig}^{\text{own}} + \sum_j x_{ig}^{\text{spill},j} + x_{ig}^{\text{bg}}$, the source allocation conditional on the total is*

$$(x_{ig}^{\text{own}}, \{x_{ig}^{\text{spill},j}\}_j, x_{ig}^{\text{bg}}) \mid x_{ig} \sim \text{Multinomial}(x_{ig}; \pi_{ig}), \quad (8)$$

with

$$\pi_{ig}^{\text{own}} = \frac{(1 - \eta_i) \ell_i \mu_{ig}}{\lambda_{ig}}, \quad \pi_{ig}^{\text{spill},j} = \frac{\eta_j \ell_j w_{j \rightarrow i} \mu_{jg}}{\lambda_{ig}}, \quad \pi_{ig}^{\text{bg}} = \frac{\nu_i \beta_g}{\lambda_{ig}}. \quad (9)$$

Proof sketch. Marginalising ξ in (7) gives the standard Gamma–Poisson (NB) marginal. Conditional on ξ_{ig} , the three sources are independent Poissons with rates $\xi_{ig}(1 - \eta_i) \ell_i \mu_{ig}$, $\xi_{ig} \eta_j \ell_j w_{j \rightarrow i} \mu_{jg}$ and $\xi_{ig} \nu_i \beta_g$; the conditional distribution of independent Poissons given their sum is Multinomial with probabilities proportional to the rates, and the shared ξ_{ig} cancels. $\square$

Remark 1. *Proposition 1 is the technical reason for the shared cell–gene over-dispersion variable. Assigning independent NB distributions to each source would make the observed total non-NB and the allocation intractable; the shared- ξ construction buys exactness at the cost of assuming the three sources share an over-dispersion mechanism, which is reasonable when over-dispersion is driven by detection efficiency and optical crowding rather than by biological source.*

Deliverables that fall out for free.

$$\tilde{x}_{ig} = \pi_{ig}^{\text{own}} x_{ig} \quad \text{decontaminated counts,} \quad (10)$$

$$\hat{x}_{ig} = \tilde{x}_{ig} + \eta_i \ell_i \mu_{ig} \quad \text{restored counts (leak-back corrected),} \quad (11)$$

$$q_i = \sum_g \tilde{x}_{ig} / \sum_g x_{ig} \quad \text{segmentation confidence} \in [0, 1]. \quad (12)$$

$\tilde{x}$ is exported as an AnnData layer; q_i is exported to `obs` and used to weight the diffusion operator (Sec. 8.2) and to flag cells for review.

4.4 Background from negative controls

$\beta \in \Delta^{G-1}$ and the per-cell scale ν_i are anchored, not free. Let $\mathcal{C}$ be the set of negative-control features and n_i^{neg} the total negative-control counts in cell i . We set $\nu_i = \kappa_\beta a_i$ and fit κ_β by matching the expected total background to the negative-control rate rescaled by the ratio of real to control probes:

$$\kappa_\beta = \frac{|\mathcal{G}|}{|\mathcal{C}|} \cdot \frac{\sum_i n_i^{\text{neg}}}{\sum_i a_i}. \quad (13)$$

β_g is initialised uniform and given a Dirichlet prior concentrated at uniform, $\beta \sim \text{Dirichlet}(\alpha_\beta \mathbf{1})$ with α_β large, so per-gene background deviations must be strongly supported. This anchoring matters: platforms differ enormously in background (reported negative-probe rates around 0.06% for Xenium versus roughly 10% for CosMx), and a model that fits $\nu_i \beta_g$ freely will, on a low-background platform, use the background term to absorb real biology. Equation (13) makes the background term as large as the controls say it is and no larger.

4.5 Spillover kernel: geometry only, by design

$$w_{j \rightarrow i} = \frac{\exp(e_{j \rightarrow i})}{\sum_{i' \in \mathcal{N}(j)} \exp(e_{j \rightarrow i'})}, \quad (14)$$

$$e_{j \rightarrow i} = -\frac{d_{ij}}{\tau} + \alpha_1 \tilde{c}_{ij} + \alpha_2 \frac{a_i^{1/D}}{a_i^{1/D} + a_j^{1/D}} + \alpha_3 \mathbb{I}[|z_i - z_j| < \epsilon_z], \quad (15)$$

where $\tilde{c}_{ij} = c_{ij} / \sum_{i'} c_{ji'}$ is the normalised shared-boundary fraction, and the last term (3D only) upweights cells in the same z -plane. The leakage fraction is

$$\eta_j = \sigma(\text{MLP}_\zeta(g_j)) \cdot \eta_{\max}, \quad \eta_{\max} = 0.6. \quad (16)$$

Assumption 1 (Geometric sufficiency of contamination). η_j and $w_{j \rightarrow i}$ depend on the data only through geometry: g_j contains $\log a_j$, circularity $4\pi a_j / \text{perim}_j^2$, solidity, nucleus/cytoplasm area ratio, neighbour count, mean and minimum neighbour distance, total shared-boundary fraction, distance to tissue edge, z -extent, and (if transcripts are available) the fraction of the cell’s molecules lying beyond the 90th percentile of radial distance from the nucleus centroid. No expression feature enters g_j or (15).

This restriction is the single most important identifiability device in the model, and it is a restriction we impose deliberately even though a more flexible expression-dependent kernel would fit better. If η or w were allowed to depend on expression, the contamination term could reproduce any pattern of neighbour–neighbour correlation, including genuine niche-driven co-expression, and the decomposition into “spillover” and “communication” would collapse. Under Assumption 1 the contamination term can only ever say “cell i ’s counts look like a geometry-weighted average of its neighbours’ profiles”, and any residual neighbour-dependent structure must be explained by Δ_{ig} .

Proposition 2 (Informal identifiability of the contamination split). *Suppose (i) Assumption 1 holds; (ii) the panel contains at least two genes g, g' and two cell types $t \neq t'$ with $\mu_g^t / \mu_{g'}^t \neq \mu_g^{t'} / \mu_{g'}^{t'}$ (type-discriminative genes exist); (iii) the geometric features g_j vary across cells of the same type and the cell-type composition of neighbourhoods varies across cells with the same g_j . Then η and $\{w_{j \rightarrow i}\}$ are identified up to the resolution of the geometric feature map, and π^{own} is identified.*

Argument. The spillover term contributes to cell i a signal proportional to the *profile of a named neighbour*, not to a generic ambient profile. Condition (ii) means neighbour profiles are distinguishable; condition (iii) provides the two independent sources of variation needed to separate “how much leaks” (a function of i and j ’s geometry) from “what leaks” (a function of j ’s type). The empirical signature the model exploits is the presence of a strongly type-specific marker of type t' inside a cell whose remaining profile is type t , in an amount that scales with $\tilde{c}_{ij}$ and $1/d_{ij}$ toward a specific t' neighbour. Where no type-discriminative genes are on the panel, or where all neighbours share the cell’s type, the split is genuinely unidentifiable and the posterior reverts to the prior — correctly, and detectably via the reported posterior variance. $\square$

Remark 2 (Honest limitation). *Proposition 2 is a structural argument, not a theorem with a full proof; conditions (ii) and (iii) are checkable empirically and we recommend reporting them as a diagnostic (fraction of cells with ≥ 1 neighbour of a different coarse type; number of markers with type specificity above a threshold). Homogeneous tissue regions are the regime where CASTLE’s contamination correction is weakest, and the model should say so rather than silently over-correct.*

4.6 Optional transcript-level refinement

If the molecule table is available, source responsibilities can be sharpened per molecule rather than per gene. For molecule t assigned to cell i at position q_t , define r_t = distance from the nucleus centroid normalised by the local cell radius, and b_{tj} = proximity to the i - j shared boundary. Model

$$p(q_t \mid \text{source}) \propto \begin{cases} \exp(-r_t^2/2\sigma_{\text{own}}^2) & \text{own,} \\ \exp(-b_{tj}^2/2\sigma_{\text{sp}}^2) & \text{spillover from } j, \\ \text{uniform over the cell} & \text{background,} \end{cases} \quad (17)$$

and multiply these densities into the Multinomial probabilities (9) before renormalising. Aggregating molecule-level responsibilities back to (i, g) gives a refined π_{ig}^{own} . This module changes no other part of the architecture; it is a plug-in that increases the effective information about contamination by roughly the log of the number of molecules per gene per cell, and it is where the largest practical accuracy gain is expected for platforms that ship molecule coordinates.

4.7 Spatial doublets

Modes (i)–(ii) are contamination; a merged segment is different in kind, because no single μ_i exists. We detect it with a within-cell likelihood-ratio test: fit a two-component version of cell i , $\mu_i^{(1)}, \mu_i^{(2)}$ with mixing ρ_i , restricted to the top principal directions of the fitted f^c manifold, and compute

$$\text{LR}_i = 2 \left[\log p(x_i \mid \mu_i^{(1)}, \mu_i^{(2)}, \rho_i) - \log p(x_i \mid \mu_i) \right]. \quad (18)$$

Cells above a calibrated threshold (calibrated by simulating merges of well-separated cells within the same dataset) are flagged, down-weighted in the likelihood by $\omega_i \in (0, 1]$, and excluded from serving as spillover *donors* (their μ_j is unreliable). In 3D data this test is complemented by a vertical-signal-integrity statistic comparing the profile across z -slices of the same segment.

5 Inference model

5.1 Unrolled purification encoder for z^c

The encoder faces a chicken-and-egg problem: z_i^c should be inferred from decontaminated counts, but decontamination requires knowing all cells' profiles. We resolve it with a K -step unrolled fixed-point encoder ($K = 2$ by default), which is amortised and differentiable end-to-end.

$$h_i^{(0)} = \text{MLP}_\varphi(\log(1 + x_i/\bar{s}), u_i), \quad (19)$$

$$\mu_i^{(k)} = \text{decoder (3)–(4) evaluated at } (\mu_c(h_i^{(k-1)}), \mu_n(\{h_j^{(k-1)}\}_{j \in \mathcal{N}(i)})), \quad (20)$$

$$\lambda_i^{(k)} = \text{Eq. (6) with } \mu^{(k)}, \eta, w, \beta, \quad (21)$$

$$\tilde{x}_i^{(k)} = \left(\frac{(1-\eta_i)\ell_i\mu_i^{(k)}}{\lambda_i^{(k)}} \right) \odot x_i, \quad (22)$$

$$h_i^{(k)} = \text{MLP}_\varphi(\log(1 + \tilde{x}_i^{(k)}/\bar{s}), u_i), \quad (23)$$

with shared weights φ across steps. The variational posterior is

$$q_\varphi(z_i^c \mid \cdot) = \mathcal{N}(\mu_c(h_i^{(K)}), \text{diag } \sigma_c^2(h_i^{(K)})). \quad (24)$$

$\bar{s}$ is a global scale constant (median total count), not a per-cell library-size normaliser: the per-cell scale is a modelled quantity ℓ_i , and dividing it out in the encoder would destroy the information the model needs.

5.2 Niche encoder for z^n : self-excluded, rotation-invariant, multi-scale

Three architectural constraints define this encoder.

(1) Self-exclusion. z_i^n is computed with the self-loop *removed*: it is a function of $\{h_j^{(K)}\}_{j \in \mathcal{N}(i) \setminus \{i\}}$ only. This is a hard constraint and is the primary defence against z^n degenerating into a copy of z^c . Note also that neighbours enter through their *purified* representations $h_j^{(K)}$, so contamination cannot manufacture niche signal — a cell that received spillover from a T cell does not thereby acquire a T-cell-rich niche embedding unless a T cell is genuinely there.

(2) Rotation and translation invariance. Tissue has no canonical orientation; a rotated section must yield identical embeddings. Absolute coordinates never enter. Radial information enters as distances; angular information enters only through the *magnitudes* of azimuthal Fourier coefficients, which are $SO(2)$ -invariant:

$$A_{i,f}^{(m)} = \left| \sum_{j \in \mathcal{N}(i) \setminus \{i\}} \varrho(d_{ij}) h_{jf}^{(K)} e^{im\theta_{ij}} \right|, \quad m \in \{0, 1, 2\}, \quad (25)$$

where θ_{ij} is the angle of $p_j - p_i$ and ϱ is a radial kernel. $m = 0$ is the neighbourhood mean; $m = 1$ is the magnitude of the local expression *gradient*, which distinguishes a cell at a tissue interface from one in a homogeneous region — information that mean-pooling destroys and that is essential for domain-boundary detection. In 3D, replace $e^{im\theta}$ with real spherical harmonics and take per-degree norms, which are $SO(3)$ -invariant.

(3) Multi-scale. The encoder is evaluated at S nested radii $R_1 < \dots < R_S$ (default $S = 3$, chosen as the $\{15, 40, 80\}$ th percentiles of the nearest-neighbour distance distribution scaled to give roughly $\{6, 20, 60\}$ neighbours). At each scale a graph-attention layer with edge features $[\varrho(d_{ij}), \tilde{c}_{ij}]$ aggregates $\{h_j\}$, and its output is concatenated with the invariants (25):

$$h_i^n = \left\|_{s=1}^S \left[\text{GAT}^{(s)}(\{h_j^{(K)}\}_{j \in \mathcal{N}_s(i)}) \parallel A_i^{(0:2)}|_{R_s} \right], \quad q(z_i^n) = \mathcal{N}(\mu_n(h_i^n), \text{diag } \sigma_n^2(h_i^n)). \quad (26)$$

Attention is used rather than mean-pooling so that a single informative neighbour is not washed out, but the network is deliberately *shallow* (one layer per scale, no stacking) because depth on a spatial graph produces over-smoothing: multi-scale breadth substitutes for depth. Nested radii additionally give a natural hierarchy of domains (fine niches nested in coarse tissue compartments) by clustering the per-scale blocks separately.

6 Objective

6.1 Graph-coupled ELBO

Because λ_{ig} in (6) depends on neighbours' μ_j , hence on z_j^c, z_j^n , the likelihood is graph-coupled. Let $\mathcal{Z} = \{z_i^c, z_i^n\}_{i=1}^N$. The evidence lower bound is

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} = & \sum_{i=1}^N \omega_i \mathbb{E}_{q(\mathcal{Z}_{\mathcal{N}[i]})} \left[\sum_{g=1}^G \log \text{NegBinomial}(x_{ig}; \lambda_{ig}, \phi_g) \right] \\ & - \beta_c \sum_i \text{KL}(q(z_i^c) \parallel p(z_i^c | u_i)) - \beta_n \sum_i \text{KL}(q(z_i^n) \parallel p(z_i^n)) - \sum_i \text{KL}(q(\ell_i) \parallel p(\ell_i | a_i)), \end{aligned} \quad (27)$$

where $\mathcal{N}[i] = \{i\} \cup \mathcal{N}(i)$ and ω_i is the doublet down-weight of Sec. 4.7.

6.2 The asymmetric information constraint

Setting $\beta_c = \beta_n = 1$ recovers a standard VAE, and the model will place *all* explanatory burden on z^c , driving $\Delta \rightarrow 0$: since z^c is encoded from the cell's own expression, it can always explain the cell's own expression, including its niche-induced component. The niche block would then be decorative.

The fix is to make intrinsic capacity expensive and niche capacity cheap:

$$\boxed{\beta_c > 1 > \beta_n} \quad (\text{defaults } \beta_c = 4, \beta_n = 0.5). \quad (28)$$

$\beta_c \text{KL}(q(z^c) \parallel p(z^c | u))$ upper-bounds (up to a constant) β_c times the mutual information $I(x; z^c)$ under the aggregate posterior, so (28) is literally a penalty on how much the intrinsic latent is allowed to know. Neighbour-correlated variation is then cheaper to encode in z^n , and the model routes it there. This asymmetry is the mechanism by which the spatially-induced component of a cell's state is attributed to the niche rather than absorbed into cell identity; it is the same principle as the asymmetric regularisation used to obtain identifiability in variational intrinsic/spatial decompositions, adapted here to a contamination-aware likelihood.

A counter-pressure is required, or Δ will absorb everything:

$$\mathcal{L}_{\text{sparse}} = \frac{1}{NG} \sum_{i,g} |\Delta_{ig}| + \frac{\gamma_{\text{grp}}}{G} \sum_g \left(\frac{1}{N} \sum_i \Delta_{ig}^2 \right)^{1/2}, \quad (29)$$

an elementwise ℓ_1 plus a group term over genes, so that only a minority of genes are declared niche-responsive. The pair $(\beta_c, \lambda_{\text{sp}})$ controls the intrinsic/niche split and is the model's most consequential hyperparameter pair; Sec. 10 gives a data-driven selection rule rather than a fixed default.

6.3 Independence and sufficiency

Independence. A biased-corrected empirical Hilbert–Schmidt Independence Criterion on a minibatch B , with RBF kernels K (on z^c) and L (on z^n) and centring matrix H :

$$\mathcal{L}_{\text{HSIC}} = \frac{1}{(|B| - 1)^2} \text{tr}(KHLH). \quad (30)$$

Bandwidths are set by the median heuristic per minibatch. HSIC (rather than an adversarial critic) is chosen because it is a differentiable closed-form statistic with no inner optimisation, which matters for reproducibility: adversarial disentanglement is the main source of seed-sensitivity in comparable models.

Remark 3 (Why exact independence is the wrong target). z^c and z^n are genuinely dependent in tissue. $\mathcal{L}_{\text{HSIC}}$ is therefore applied with a small weight and is intended to break the degenerate solution (z^n a deterministic function of z^c), not to enforce statistical independence. We report the achieved HSIC and the normalised cross-predictability $R^2(z^c \rightarrow z^n)$ as diagnostics, and target the latter in the range 0.2–0.6: near 1 means degeneracy, near 0 means the niche block has been sterilised of real tissue structure.

Niche sufficiency. To stop z^n collapsing (the opposite failure), an auxiliary head must reconstruct the neighbourhood’s aggregate purified profile from z_i^n alone:

$$\mathcal{L}_{\text{niche}} = \sum_i \left\| \psi(z_i^n) - \sum_{j \in \mathcal{N}(i)} \frac{\tilde{x}_j}{\|\tilde{x}_j\|_1} / |\mathcal{N}(i)| \right\|_2^2. \quad (31)$$

6.4 Local isometry, for trajectory validity

Clustering and trajectory inference make opposite demands on the latent. We do *not* use a mixture-of-Gaussians prior (which would manufacture discreteness and corrupt pseudotime); instead we keep the unimodal prior (1) and add a weak local-isometry term that preserves local distances of the *purified* expression manifold:

$$\mathcal{L}_{\text{iso}} = \sum_{(i,i') \in \mathcal{M}} \left(\left\| \mu_c(h_i) - \mu_c(h_{i'}) \right\|_2 - \varsigma \cdot d_{\text{expr}}(\tilde{x}_i, \tilde{x}_{i'}) \right)^2, \quad (32)$$

where $\mathcal{M}$ is a set of pairs sampled from the k NN graph in purified expression space (so only *local* distances are constrained — global isometry is neither achievable nor desirable), d_{expr} is correlation distance on $\log(1 + \tilde{x})$, and ς is a learned global scale. Discreteness, where it exists, then emerges from the data rather than from the prior.

6.5 Full objective

$$\min_{\theta, \varphi, \omega, \zeta, \alpha, \phi, \beta} -\mathcal{L}_{\text{ELBO}} + \lambda_{\text{sp}} \mathcal{L}_{\text{sparse}} + \lambda_{\text{h}} \mathcal{L}_{\text{HSIC}} + \lambda_{\text{n}} \mathcal{L}_{\text{niche}} + \lambda_{\text{iso}} \mathcal{L}_{\text{iso}} + \lambda_{\beta} \mathcal{L}_{\beta} \quad (33)$$

where $\mathcal{L}_{\beta} = -\log \text{Dirichlet}(\beta; \alpha_{\beta} \mathbf{1})$ anchors the background profile.

7 Optimisation and scalability

7.1 Two-hop neighbour sampling

The likelihood of cell i requires μ_j for $j \in \mathcal{N}(i)$, and μ_j requires z_j^n , which requires h_k for $k \in \mathcal{N}(j)$. Training therefore samples 2-hop induced subgraphs. With a target-node batch of B cells and mean degree $\bar{k}$, the subgraph has $O(B\bar{k}^2)$ nodes; at $B = 1024$, $\bar{k} = 8$ this is $\sim 6.5 \times 10^4$ nodes, comfortably within one GPU. Gradients are taken only with respect to the loss on the B target nodes to avoid over-weighting high-degree cells.

7.2 Curriculum

A three-phase schedule is used because the contamination and disentanglement components are mutually confounded at initialisation.

Algorithm 1 CASTLE training

- 1: **Input:** X, p, a , masks, negative controls
 - 2: Build spatial graph (pruned Delaunay $\cup$ radius); compute c_{ij} , geometric features g_i
 - 3: Estimate κ_β from negative controls by (13); initialise β uniform
Phase 1 — warm-up (10% of epochs): freeze $\eta \equiv 0, \Delta \equiv 0$
 - 4: **for** minibatch **do**
 - 5: fit f^c, ϕ, ℓ as a plain NB-VAE on raw counts $\Rightarrow$ initialise the expression manifold
 - 6: **end for**
Phase 2 — contamination (40%): unfreeze η, w ; keep $\Delta \equiv 0$
 - 7: **for** minibatch **do**
 - 8: K -step unrolled encoder (19)–(22); optimise (27) $+\lambda_\beta \mathcal{L}_\beta$
 - 9: **end for**
 - 10: Run doublet test (18); set weights ω_i ; exclude flagged cells as donors
Phase 3 — joint (50%): unfreeze Δ , niche encoder, all penalties
 - 11: **for** minibatch **do**
 - 12: sample 2-hop subgraph; compute (33); Adam step; anneal β_c from 1 to 4
 - 13: **end for**
 - 14: **Output:** $Z = [z^c, z^n]$, $\tilde{X}, q, \Delta, \eta, w, \pi$, posterior variances
-

Phase 1 is necessary because contamination is only detectable once the model knows roughly what cell profiles look like; Phase 2 before Phase 3 is necessary because otherwise Δ (niche effect) and the spillover term compete to explain the same neighbour-driven correlation, and the model has no reason to prefer the correct attribution. Fitting contamination *first*, with Δ clamped to zero, lets the geometry-only kernel claim exactly the variance it can explain; whatever neighbour-dependent structure survives is then available to Δ . This ordering is a substantive design choice, and reversing it is one of the ablations.

7.3 Cost

Per epoch the dominant cost is $O(N\bar{k}KG_{\text{eff}})$ dense-decoder work. For $N = 5 \times 10^5$, $G = 480$, $\bar{k} = 8$, $K = 2$, $d_c = 20$, $d_n = 20$, a single modern GPU should complete training in a few hours; memory is dominated by the 2-hop subgraph decoder activations and is controlled by B . These are design targets from operation counts, not measured benchmarks; they must be verified empirically before being quoted.

8 Downstream: one model, four analyses

8.1 Cell type clustering

Cluster z^c (Leiden on a k NN graph in z^c , or a Gaussian mixture with stability-based model selection). Because z^n carries the context and β_c suppresses spatially-driven variance in z^c , clusters are cell types rather than location-confounded hybrids. Marker detection uses the purified counts $\tilde{x}$, which removes the most common artefact in these platforms: a marker of type t' appearing “expressed” in type t because t cells sit next to t' cells. Cells with low confidence q_i or high doublet LR_i are reported as a distinct low-confidence set rather than being force-assigned.

8.2 Trajectory inference

Pseudotime is computed on z^c using a diffusion operator whose edge weights are down-weighted by segmentation confidence:

$$W_{ii'} = \exp\left(-\frac{\|z_i^c - z_{i'}^c\|^2}{2\sigma_i\sigma_{i'}}\right) \cdot \sqrt{q_i q_{i'}}, \quad (34)$$

followed by standard diffusion-map / Palantir machinery. Two things are gained. First, the manifold is built on niche-decoupled coordinates, which addresses the standing objection that spatial pseudotime is confounded by spatial autocorrelation: a gradient across the tissue no longer masquerades as a differentiation axis, because that gradient lives in z^n . Second, contaminated cells — which otherwise appear as spurious intermediate states bridging two distinct types and create phantom trajectories — are explicitly discounted.

A *separate* spatial-gradient analysis is available by computing pseudotime on z^n , but it must be reported and interpreted as a spatial-organisation axis, not a temporal one. We recommend the software refuse to label the latter “pseudotime”.

8.3 Niche and domain discovery

Cluster z^n for niches. The multi-scale construction yields a hierarchy: clustering the R_1 block gives fine cellular neighbourhoods, the R_S block gives coarse tissue compartments, and the nesting relation between them is read off directly. The $m=1$ azimuthal invariants make domain *boundaries* detectable as high-gradient cells, which mean-pooling methods systematically blur. Niche identity can be characterised without a separate model by the mean Δ_{ig} within each niche (the genes the niche induces) and by the composition of z^c -clusters within it.

8.4 Cell–cell communication with a spillover-aware null

This is the component where contamination modelling pays off most directly. Adjacent-cell ligand–receptor co-expression is exactly the statistic that transcript spillover inflates: if some of cell i ’s counts are actually cell j ’s, then i and j will appear to co-express interacting partners regardless of biology. CASTLE provides three coupled estimators.

(a) Spatial effect attribution. Δ_{ig} is already a per-cell estimate of niche-induced expression. For a receptor R , $\Delta_{iR} > 0$ says the niche upregulates R in this cell. Regressing Δ_{iR} on neighbouring ligand abundance,

$$\Delta_{iR} \sim \gamma_{LR} \cdot \sum_{j \in \mathcal{N}(i)} \varrho(d_{ij}) \tilde{x}_{jL} + \text{controls}, \quad (35)$$

with controls for z_i^c (cell type), q_i , a_i and local density, gives a directed, effect-size estimate $\hat{\gamma}_{LR}$ rather than a co-occurrence score.

(b) Program activities. If the masked head (5) is used, the program coordinates of z^n are named ligand–receptor/pathway activities, usable directly.

(c) The spillover-aware null. The key inferential device. Generate M synthetic datasets from the *fitted* model with the niche effect ablated:

$$X^{(m)} \sim p_\theta(X \mid z^c, \Delta \equiv 0, \hat{\eta}, \hat{w}, \hat{\beta}, \text{geometry}), \quad m = 1, \dots, M. \quad (36)$$

These data contain the true cell types, the true tissue architecture, and the true fitted contamination — but no communication. Any test statistic T (a CellChat-style score, a spatial co-expression statistic, or $\hat{\gamma}_{LR}$ from (35)) is computed on the real data and on each $X^{(m)}$, and significance is

$$p_{LR} = \frac{1 + \#\{m : T(X^{(m)}) \geq T(X)\}}{M + 1}. \quad (37)$$

This directly answers the question the field’s standard permutation nulls cannot: *is this ligand–receptor signal larger than what segmentation error alone would produce in this tissue with these cell types?* Label-permutation nulls destroy tissue architecture and therefore have no contamination structure to compare against; (36) preserves it exactly.

Remark 4 (What this does and does not establish). *Passing the null in (37) rules out contamination and architecture as sufficient explanations. It does not establish that a ligand–receptor interaction is functionally occurring. Panel coverage of curated ligand–receptor pairs is limited (often heavily so for < 500-gene panels), the statistic remains correlational, and no expression-based method can substitute for perturbation evidence. The correct claim is “this co-expression pattern is not explained by segmentation error”, which is a strictly stronger and more honest claim than current practice, but not a causal one.*

9 Uncertainty quantification

Four calibrated quantities are exported. (i) Posterior variance $\sigma_c^2(i)$, $\sigma_n^2(i)$ — amortised, so under-dispersed; calibrate on held-out genes by checking coverage of predictive intervals. (ii) Segmentation confidence q_i (12). (iii) Doublet statistic LR_i (18), with a threshold calibrated by in-dataset simulated merges. (iv) Cluster-assignment confidence via *split conformal prediction*: hold out a calibration set, use $1 - \hat{p}(\text{assigned label})$ as the conformity score, and emit prediction *sets* at level $1 - \alpha$. Cells receiving a set of size > 1 are ambiguous and are reported as such. This is cheap, distribution-free, and directly usable by an experimentalist deciding which annotations to trust.

10 Hyperparameters and defaults

Selecting (β_c, λ_{sp}) without ground truth. These control the intrinsic/niche split and should not be fixed blindly. We select them on a *held-out-gene* criterion: hold out 10% of panel genes from the encoder input (but not the likelihood), and choose the pair maximising held-out predictive log-likelihood, subject to the diagnostic constraint $R^2(z^c \rightarrow z^n) \in [0.2, 0.6]$. This makes the choice data-driven and falsifiable, and it does not use the clustering labels that the model will later be evaluated on — avoiding the circularity that pervades spatial-embedding benchmarking.

Table 2: Defaults. G = panel size.

Symbol	Default	Note
d_c	$\max(10, \min(30, \lceil G/25 \rceil))$	intrinsic dimension
d_n	d_c	niche dimension
K	2	unrolled purification steps; $K=3$ gives $< 1\%$ change typically
$S, \{R_s\}$	3	radii giving $\approx \{6, 20, 60\}$ neighbours
$\bar{k}$	8	mean degree of R_1 graph
$\eta_{\max}$	0.6	hard cap on leakage fraction
β_c	4 (annealed $1 \rightarrow 4$)	intrinsic information cost
β_n	0.5	niche information cost
λ_{sp}	selected (below)	sparsity on Δ
λ_h	10^{-2}	HSIC weight; small by design
λ_n	1	niche sufficiency
λ_{iso}	10^{-1}	local isometry
α_β	10^3	Dirichlet concentration anchoring background
B	1024 target nodes	2-hop sampled subgraph
optimiser	Adam, 10^{-3}	cosine decay, gradient clip 5

11 Evaluation, ablations, and decision thresholds

11.1 Ground-truth strategies

Real ground truth is scarce, so we use four complementary sources, none of which is circular.

1. **Semi-synthetic contamination.** Take a real dataset, fit CASTLE, then *simulate* additional known spillover by moving a known fraction of transcripts between adjacent cells. Measure recovery of η , w , and π^{own} (correlation, calibration curves), and the recovery of the pre-contamination embedding.
2. **Nucleus-only vs. expanded segmentation.** Segment the same slide two ways. Nucleus-only counts are low-contamination, low-power; expanded counts are high-contamination, high-power. A correct model fitted to expanded counts should recover profiles closer to the nucleus-only profiles than raw expanded counts are. This uses no external data and is available for every Xenium/CosMx dataset.
3. **Matched dissociated reference.** Where matched scRNA-seq/snRNA-seq exists, purified per-cell-type profiles $\tilde{\mu}$ should correlate better with the dissociated reference than raw profiles do, restricted to panel genes.
4. **Anatomical annotation.** For brain data with atlas registration, anatomical regions provide domain labels not derived from expression clustering.

11.2 Baselines and metrics

Baselines: PCA and scVI on raw counts (non-spatial); BANKSY; CellCharter; STAGATE; GraphST; a disentangling spatial VAE (SIMVI-class); NicheCompass; and, critically, **scVI/PCA applied to counts pre-corrected by a dedicated decontamination tool** — this is the sharpest baseline, because it tests whether joint modelling beats sequential correct-then-embed.

Metrics: ARI/NMI against expert annotation (clustering); CHAOS/PAS spatial coherence and ARI against anatomical labels (domains); contamination-recovery correlation and doublet AUROC (segmentation); held-out gene predictive log-likelihood (fit); trajectory correlation with known axes and phantom-branch rate; ligand-receptor recovery under the null of (36), plus false-positive rate on *simulated* contamination-only data (a method that reports significant communication on data generated with $\Delta \equiv 0$ is broken); and seed-stability across ≥ 10 seeds.

11.3 Ablations

Each removes one design decision: (1) no contamination term (raw NB likelihood); (2) contamination as preprocessing rather than joint; (3) expression-dependent spillover kernel (violating Assumption 1); (4) free background versus negative-control anchoring; (5) $\beta_c = \beta_n = 1$ (no asymmetry); (6) no HSIC; (7) niche encoder with self-loop included; (8) mean-pooling instead of azimuthal invariants; (9) single-scale niche; (10) no local isometry; (11) mixture prior instead of Gaussian; (12) $K = 1$ (no unrolling); (13) Phase 2/3 order reversed; (14) k NN instead of pruned Delaunay $\cup$ radius; (15) no transcript-level module.

11.4 Decision thresholds (when to abandon parts of this design)

Stated in advance to keep the evaluation honest.

- If ablation (2) — sequential decontamination then scVI — matches joint modelling within noise on clustering, trajectory *and* communication metrics, then the joint likelihood is not earning its complexity, and CASTLE should be reduced to a decontamination tool plus a standard VAE.
- If ablation (5) shows the intrinsic/niche split is insensitive to β_c , the disentanglement is not identified in practice; report Z as a single block and drop the causal language about spatial effects.
- If contamination correction reduces median per-cell counts by $> 35\%$ without improving any ground-truth metric in Sec. 7.1, the model is over-correcting; cap $\eta_{\max}$ downward or disable.
- If z^c -clustering does not beat PCA on raw counts by a clear margin on ARI, the entire generative apparatus is not justified for clustering, and the honest recommendation is PCA for clustering plus CASTLE only for contamination and communication.

12 Assumptions, failure modes, and limitations

Assumption / risk		Consequence if violated	Mitigation / detection
Contamination geometry-predictable (Assumption 1)	is	Systematic, gene-dependent probe artefacts misattributed	Residual diagnostics of π^{own} against gene properties; report unexplained neighbour correlation
Type-discriminative genes on panel (Prop. 2)		Contamination split unidentified in homogeneous tissue	Report per-region identifiability diagnostic; posterior reverts to prior and variance is reported
Shared over-dispersion across sources (Prop. 1)		Mis-calibrated tails	Posterior predictive checks on count distributions

Assumption / risk		Consequence if violated	Mitigation / detection
Additive decomposition (3)	log-	Multiplicative/interaction niche effects mis-modelled	f^n conditions on z^c , capturing type-specific response; check residual interaction
β_c controls the split		Arbitrary intrinsic/niche attribution	Held-out-gene selection (Sec. 10); sensitivity curve always reported
Single μ_i per segment		Merged cells break the model	Doublet LR test, down-weighting, donor exclusion
Over-correction of counts		Loss of statistical power in DE	Report count reduction; cap $\eta_{\max}$; decision threshold above
2D model of 3D tissue		Systematic z -direction spillover missed	3D features and z -plane terms when available; otherwise state as a known bias
Spatial pseudotime		Spatial gradient read as time	Software refuses “pseudotime” label for z^n -based orderings
Communication is correlational		Over-claiming	Null (36) only rules out contamination; report panel coverage of each LR pair
Graph construction		Results depend on $\mathcal{N}$	Ablation (14); ship sensitivity analysis by default
Amortised posteriors		Under-dispersed uncertainty	Conformal calibration for labels; coverage checks

Two limitations worth stating plainly. First, CASTLE now fits one dataset at a time by design; multi-sample studies must integrate z^c post hoc (e.g. Harmony/scArches on z^c), and cross-sample niche comparison is not addressed here. Second, the model’s central promise — that jointly modelling contamination improves the embedding rather than merely producing a cleaner count matrix — is a hypothesis. Ablation (2) is designed to falsify it, and the decision thresholds above commit in advance to what happens if it fails.

13 Implementation notes

Target stack: PyTorch + PyTorch Geometric, with `scvi-tools`-style module structure; AnnData/ SpatialData I/O; `squidpy` for graph construction and c_{ij} from polygons; outputs written as `obsm['X_castle_c']`, `obsm['X_castle_n']`, `layers['purified']`, `layers['spatial_effect']`, `obs['seg_confidence']`, `obs['doublet_lr']`, and `uns['ccc']`. The molecule-level module reads the platform transcript tables directly. A CPU-only fallback path should exist for $N < 10^5$, since many labs generating these data do not have GPU access, and a method intended as a default first analysis must run without one.

This document specifies a design, not a validated method. Every quantitative claim about performance is a target to be tested by the protocol in Sec. 11, and the decision thresholds in Sec. 7.4 exist so that a negative result leads to simplification rather than to post-hoc rescue.